

Introduction

This project focuses on solving a common problem faced by students living in shared accommodations: managing and splitting expenses. In daily life, expenses such as food, rent, and utilities are often paid by different individuals, making it difficult to track contributions accurately. Existing solutions are either too complex or require constant internet access. This project aims to provide a simple, offline-friendly solution using Python to manage shared expenses efficiently.

Task 1 – Problem Identification

The problem was identified through personal observation in hostel life. Students often face confusion regarding shared expenses, especially when multiple people are involved. Manual tracking leads to errors and conflicts. This highlighted the need for a simple and reliable system to manage expenses.

Task 2 – Approach

The solution uses a command-line interface where users can input expenses, payer details, and participants. The system stores data in a JSON file and calculates balances dynamically. The logic ensures equal distribution of expenses among participants.

Task 3 – Implementation

The project is implemented using Python with modular design. Separate files handle logic, data storage, and user interaction. JSON is used for persistent storage. The system calculates net balances based on contributions and expenses.

Task 4 – Challenges

One challenge was designing a simple yet accurate balance calculation system. Handling user input and ensuring correct data formatting was also a difficulty. Debugging logic errors in balance calculation required multiple test cases.

Task 5 – Learning

This project improved understanding of file handling, modular programming, and real-world problem solving. It also helped in structuring code logically and improving debugging skills.

Conclusion

The project successfully demonstrates a practical solution to a real-world problem using simple programming techniques. It highlights the importance of clarity and usability over complexity.